

EDUARDO DE SOUZA MARQUES

AI SAFETY & AGENT-GOVERNANCE ENGINEER · RED-TEAMING · AUTONOMOUS SYSTEMS

Minas Gerais, Brazil (Remote) · English (Fluent) / Portuguese (Native) · [linkedin.com/in/edusouzamarques](#) · [github.com/edusouzamarques](#)

SUMMARY

Applied AI Systems Engineer specializing in the reliability, runtime safety, and governance of autonomous LLM multi-agent systems. Author of 14 production open-source Python libraries with 3,580+ automated unit/regression tests and 100% green CI across Linux, macOS, and Windows. Expertise covers fail-closed boundary enforcement, multi-model adversarial juries, audit logging, and token cost containment. Combined with 13.5 years of high-consequence operational discipline in public safety (PMMG), bringing an adversarial mindset to red-teaming AI models, hunting jailbreaks, mitigating prompt injection, and ensuring systematic policy compliance.

TECHNICAL COMPETENCIES

Core: AI Safety · LLM Red-Teaming · Adversarial Prompt Testing · Jailbreak Hunting · Guardrails · Autonomous Agent Governance · Failure Taxonomies · Python · Multi-Model Consensus · Test-Driven Development (pytest, CI/CD) · OWASP Top 10 for LLMs · Deterministic Error Handling · Cryptographic Provenance.

OPEN-SOURCE PORTFOLIO — AUTHOR (14 LIBRARIES · 3,580+ TESTS · CI GREEN)

deputy — Bounded delegation engine for autonomous agents: enforces reserved execution zones no quorum can unlock, caps consecutive self-authorizations, and provides tamper-proof activity journaling.

devil-advocate — Adversarial review chamber forcing cross-vendor LLMs to attack, stress-test, and refute proposed actions before physical execution.

agent-gates — Deterministic, fail-closed runtime precondition framework intercepting agent tool-calling; blocks unauthorized shell and file operations.

polycourt — Weighted multi-LLM adjudication engine with quorum thresholds and circuit-breakers; drastically mitigates single-model hallucination risks.

spend-guard · **gpu-spot-guard** — FinOps hard-stop budget ceilings, token limiters, and automated cloud GPU lifecycle reclamation.

error-registry — Structured error taxonomy and institutional memory preventing agents from repeating known execution failures.

asset-ledger · **durable-context** · **lane-router** · **mcp-navigator** · **steady-driver** · **video-assembly-kit** · **comfy-ui2api** — Deterministic media provenance, compaction-resistant memory, Model Context Protocol (MCP) tooling, cost-aware LLM routing, and headless web drivers.

PROFESSIONAL EXPERIENCE

AI Safety & Agent Systems Engineer — Pixel Pro Ads / Open Source

May 2024 – Present

- Architected and released 14 open-source Python libraries with ~3,580 automated tests focused on AI guardrails, model evaluation, and runtime containment.
- Developed multi-agent red-teaming pipelines testing cross-vendor LLMs (Claude, GPT, DeepSeek, Qwen) for prompt injection and policy compliance.
- Implemented deterministic pre-execution gates and budget limits protecting production agent workflows from runaway cost and infinite loops.

Digital Systems & Media Automation Lead — Pixel Pro Ads Ltda

2021 – 2024

- Built high-throughput automated media assembly pipelines integrating Python, FFmpeg, speech synthesis, and cloud GPU rendering APIs.
- Led operational workflows, direct-response infrastructure, and programmatic content deployment systems.

Public Safety & Operations Specialist / Corporal — Polícia Militar de Minas Gerais (PMMG)

2006 – 2019 (13.5 yrs)

- Executed frontline public safety, threat modeling, operational risk mitigation, and compliance enforcement under high-stress conditions.
- Managed strict chain of custody, evidence preservation, and objective investigative documentation for criminal judicial proceedings.
- Awarded internal commendations for tactical discipline; resigned honorably in 2019 to transition into technical engineering.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (Tecnólogo) in International Trade — Higher Education Degree	2012
Technical Degree in Public Safety — Minas Gerais Military Police Academy	2006
Cisco Certified Track: Python Essentials 1 · Networking Basics · Introduction to Cybersecurity — Cisco NetAcad	2026